Checking Arbitrage in a Finite Option Set

Theory, finite-grid proofs, LP formulations, and Python checks

Mike DeLL IbeX

April 28, 2026

Why finite option sets matter

Observed market data

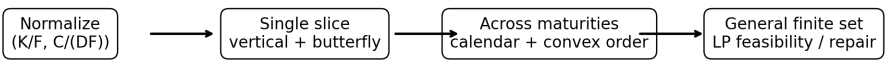
Real screens give a finite set of quotes:

- finitely many strikes;
- finitely many maturities;
- missing nodes and stale prices;
- sometimes only bids and asks.

The first question

Does the finite quote set admit *any* arbitrage-free model before interpolation or calibration?

Recommended finite-set arbitrage-checking workflow



Normalize first

Forward-discount normalization

For maturity T_j with discount factor D_j and forward F_j:

$$k = \frac{K}{F_j}, \qquad c_j(k) = \frac{C_j(K)}{D_j F_j}.$$

If $M_j = S_{T_j}/F_j$, then

$$c_j(k) = \mathbb{E}[(M_j - k)^+], \qquad \mathbb{E}[M_j] = 1.$$

Key payoff of normalization

Cross-maturity comparisons should be made in normalized coordinates. Calendar arbitrage is a statement about $c_j(k)$, not raw prices $C_j(K)$ unless the grids are adjusted correctly.

Single maturity: the whole problem is in the slopes

With $k_0 = 0$, $c_0 = 1$, and

$$q_i = \frac{c_{i-1} - c_i}{k_i - k_{i-1}}, \qquad i = 1, \dots, n,$$

one-slice no-arbitrage is equivalent to

$$1 \geq q_1 \geq q_2 \geq \cdots \geq q_n \geq 0,$$

plus exclusion of the edge case $c_n > 0$ with $q_n = 0$.

Vertical spreads

$$0 \leq \frac{(M - k_{i-1})^+ - (M - k_i)^+}{k_i - k_{i-1}} \leq 1$$

therefore $0 \leq q_i \leq 1$.

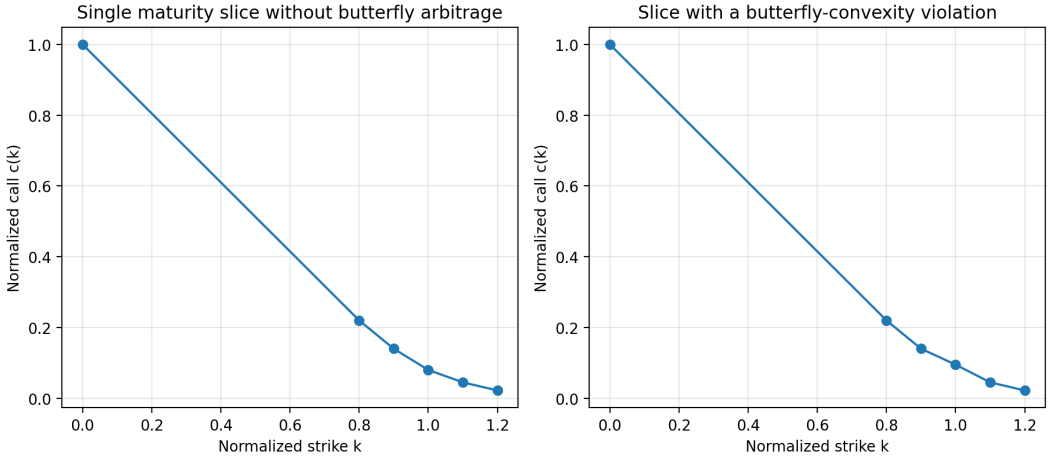
Butterflies

Long butterfly payoffs are nonnegative, hence

$$q_i - q_{i+1} \geq 0.$$

This is exactly discrete convexity in strike.

Butterfly arbitrage is discrete non-convexity



Left: convex decreasing normalized calls. Right: the middle quote is too high, so the butterfly cost becomes negative.

Constructive proof: finite-support model from slopes

If

$$1 \geq q_1 \geq \cdots \geq q_n \geq \alpha \geq 0,$$

define $x_* = k_n + c_n/\alpha$ and the masses

$$p_0 = 1 - q_1, \qquad p_i = q_i - q_{i+1} \ (i = 1, \ldots, n-1), \qquad p_n = q_n - \alpha, \qquad p_* = \alpha.$$

Consequences

- the masses are nonnegative and sum to one;
- the resulting discrete law has mean one;
- its piecewise-linear call function matches the quoted slice exactly.

Edge case

If $c_n > 0$ but $q_n = 0$, no exact model exists: convexity would force the extension to stay flat forever.

Common normalized grid across maturities

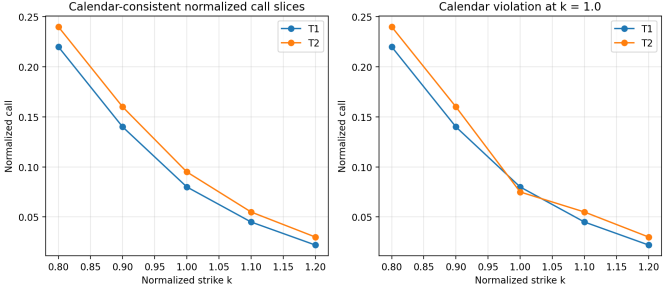
For maturities $T_j < T_{j+1}$ on the same normalized strike grid,

$$c_{i,j+1} - c_{i,j} \geq 0$$

is the finite-set calendar condition.

Why it is exact

After normalization, every slice has mean one. These inequalities imply convex-order growth across maturities, so Strassen's theorem yields a martingale coupling.



Absolute-price form of the calendar test

If the quoted strikes satisfy $K_{i,j} = k_i F_j$, then

$$c_{i,j} = \frac{C_{i,j}}{D_j F_j}, \qquad c_{i,j+1} \geq c_{i,j} \iff C_{i,j+1} - \frac{D_{j+1} F_{j+1}}{D_j F_j} C_{i,j} \geq 0.$$

Practical meaning

Equal absolute strikes are not the right comparison in general. Compare the same normalized strike K/F_T, or equivalently the same log-moneyness.

Common implementation mistake

A surface can look monotone in raw call prices and still fail calendar no-arbitrage once forwards and discounting are handled correctly.

Irregular finite sets: solve a linear program

Choose support nodes $x_{j,a}$ at each maturity. Solve for marginal masses $u_{j,a}$ and coupling masses $p_{j,a,b}$.

$$u_{j,a} \geq 0, \quad \sum_a u_{j,a} = 1, \quad \sum_a x_{j,a} u_{j,a} = 1,$$

$$\sum_a (x_{j,a} - k_{j,\ell})^+ u_{j,a} = c_{j,\ell},$$

$$p_{j,a,b} \geq 0, \quad \sum_b p_{j,a,b} = u_{j,a}, \quad \sum_a p_{j,a,b} = u_{j+1,b},$$

$$\sum_b x_{j+1,b} p_{j,a,b} = x_{j,a} u_{j,a}.$$

Why LP is the natural tool

Missing strikes, mixed grids, bid-ask intervals, and quote repair all fit into one feasibility / optimization framework.

Python examples for butterfly and calendar checks

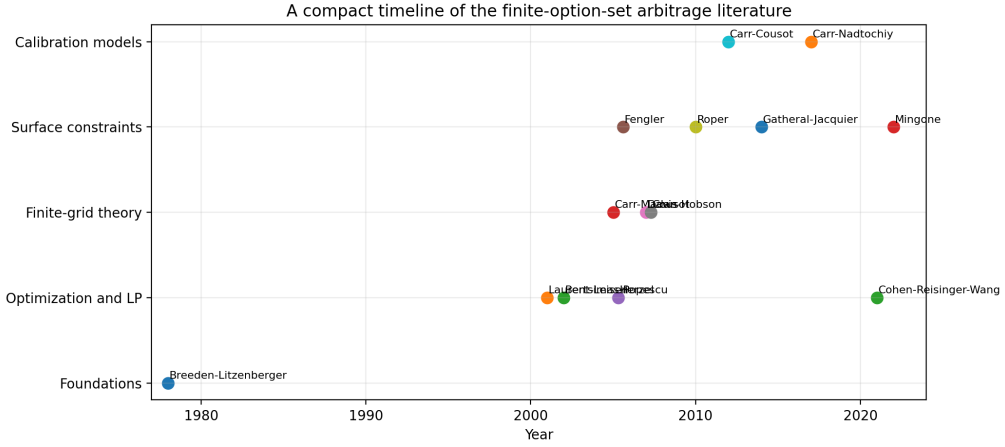
```
def check_vertical_and_butterfly(normalized_strikes, normalized_calls):
    k = np.asarray(normalized_strikes, dtype=float)
    c = np.asarray(normalized_calls, dtype=float)
    k_aug = np.concatenate(([0.0], k))
    c_aug = np.concatenate(([1.0], c))
    slopes = np.diff(c_aug) * (-1.0) / np.diff(k_aug)
    return slopes, np.all(slopes >= 0), np.all(np.diff(slopes) <= 0)


def check_calendar_same_normalised_strikes(grid):
    diffs = np.diff(np.vstack(grid), axis=0)
    return diffs, np.all(diffs >= 0)
```

Example outputs

Good slice slopes: $(0.975, 0.800, 0.600, 0.350, 0.230)$. Bad butterfly cost: -0.05.
Calendar violation example: differences $(0.02, 0.02, -0.005, 0.01, 0.008)$.

Literature map



Exact finite-set theory: Carr-Madan (2005), Davis-Hobson (2007), Cousot (2007).
Optimization and repair: Bertsimas-Popescu (2002), Herzel (2005), Cohen-Reisinger-Wang (2021).
Surface constraints: Fengler (2009), Gatheral-Jacquier (2014), Mingone (2022), Wystup (2020).

From finite quotes to arbitrage-free surfaces

Practitioner models often work with total variance $w(k, T) = \sigma_{BS}^2(k, T)T$.

Surface-level calendar condition

$$\partial_T w(k, T) \geq 0.$$

Surface-level butterfly condition (Gatheral-Jacquier)

$$g(k) = \left(1 - \frac{kw'(k)}{2w(k)}\right)^2 - \frac{w'(k)^2}{4}\left(\frac{1}{w(k)} + \frac{1}{4}\right) + \frac{w''(k)}{2} \geq 0.$$

Important

These are downstream conditions. First clean the finite quote set. Then fit SVI, eSSVI, splines, or a calibrated model.

Takeaways

1. Normalize by forwards and discounts before comparing maturities.
2. On one slice, check monotonicity plus discrete convexity of call prices in strike.
3. On a common normalized grid, calendar no-arbitrage is monotonicity in maturity of normalized calls.
4. On irregular grids, solve a martingale-feasibility LP or a quote-repair LP.
5. Only after the finite quote set is clean should one interpolate or fit a smooth surface.

Finite-set arbitrage checks are not a preprocessing nuisance.
They are the mathematically correct first layer of the problem.